

closure_evidence_20260923

BOB-156 closure evidence — 2026-09-23

Root cause + fix

The BOB-145 event-loop-blocking regression guard measured `worst_gap` (max wall-clock heartbeat gap) against a fixed absolute ceiling (900ms→1500ms). `merge_results` never yields, so `worst_gap` $\approx$ `merge_wall` — the ceiling was effectively an absolute wall-clock bound on `merge_results`'s duration. Empirically measured on this host: wall-clock time for this pure-CPU code inflates dramatically under contention from UNRELATED processes (up to ~20x observed at 7.5x/core oversubscription), while `time.process_time()` (actual CPU-seconds consumed) does not — the OS scheduler time-slices the process among more competitors, stretching wall-clock without changing the CPU work actually done. The guard was measuring host business, not code regression — the exact “weather report” symptom.

Fix: `time.process_time()`-measured CPU time is now the PRIMARY regression gate (`MAX_MERGE_CPU_S` = 1.5, same margin logic as the original design — >7x above worst observed contended CPU, >2x below the pre-fix regression's CPU). The heartbeat/`worst_gap` mechanism is retained as diagnostic evidence plus a generous 15s wall-clock backstop that only fires on a genuinely catastrophic freeze (BOB-137 measured minutes), never on ordinary contention. No production code changed — this was a measurement-methodology defect in the test, not a code regression; `deduplicator.py` confirmed byte-identical (sha256 89b20304...df3cf8, matching the hash cited in the original BOB-145 fix commit) before and after.

Independently re-verified this session (coordinator, including a

genuine paired-mutation revert/restore against the real pre-fix commit)

```
$ .venv/bin/python -m pytest tests/unit/merge_service/  
test_dedup_event_loop_blocking.py -v --import-mode=importlib  
... 7 passed in 1.01s
```

```
$ sha256sum download-proxy/src/merge_service/deduplicator.py  
89b203043e29584a565b34ca71af2f16e399ab7e1f1e011bbdb6e41a94df3cf8  
(unchanged)
```

```
$ git show 0572b71^:download-proxy/src/merge_service/  
deduplicator.py > deduplicator.py # revert to pre-BOB-145  
$ .venv/bin/python -m pytest tests/unit/merge_service/  
test_dedup_event_loop_blocking.py::TestMergeResultsDoesNotBlockThe  
-v --import-mode=importlib  
FAILED -- AssertionError: merge consumed 5551.8ms of CPU time  
(ceiling 1500ms) --  
    this is a regression in the work merge_results does, not host  
contention  
    ... This is the BOB-145 signature: while a merge this expensive  
runs,  
    port 7187 answers nothing (BOB-137).
```

```
$ (restored fixed deduplicator.py, sha256 confirmed byte-  
identical)  
$ .venv/bin/python -m pytest tests/unit/merge_service/  
test_dedup_event_loop_blocking.py -q --import-mode=importlib  
7 passed in 1.06s
```

Genuine paired-mutation confirmed independently: reverting the real code that BOB-145 fixed reproduces the exact CPU-time-based failure signature the new guard is designed to catch; restoring returns to green.

Honest boundary (not silenced, per the subagent's own report)

CPU-time-as-oracle has a measured, stated limit (~1.3-1.9x overhead even under extreme contention, from cache/context-switch effects) — the chosen 1.5s ceiling has ample margin either side of this. Robustness across genuinely different/slower HARDWARE (vs. contention

specifically on this one 16-core host) was argued by analogy, not independently measured on a second machine (none available in scope).

git diff -stat

```
tests/unit/merge_service/test_dedup_event_loop_blocking.py | 155 +
+++++-----
1 file changed, 138 insertions(+), 17 deletions(-)
```